# Electric car: launch the design

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Plan a safe team workflow and identify the car's energy pathway.

**SUCCESS CRITERIA - I can...**

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use "prototype" and "constraint" to explain your model or result. Sentence starters are optional.

**prototype**: a version built to test a design

**constraint**: a requirement or limit that the design must meet

**evidence**: observations or measurements that support an idea

## 01 Bring back an earlier idea

A stationary coil has constant magnetic flux 0.003 Wb through each turn. Does that constant flux produce sustained induced emf?
**Explain**.

**Word help:** flux = a measure of magnetic field passing through a chosen area

**Sentence starter:** My claim is … . The diagram/data show …, so … .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** average speed = distance/elapsed time.
Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

**Worked reasoning:** A design claim becomes testable when it names a measurement. "The car is fast" can become "The car travels 3 m in under 2 s on this level track." That condition corresponds to average speed above 1.5 m/s.

**Your turn: Draw** a test track with start, finish, distance and timing method labeled.

**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** ① ② ③

## 03 Look and explain

**Sketch** your proposed car.
**Label** the battery, switch, motor, driven axle and wheels.
Add a theme that will not obstruct moving parts.

**Sentence starter:** The … in my model represents … . Its label shows … .




Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

**Marzano check - circle:** ① ② ③

## 04 Build the solution

Assign a build lead, safety checker and recorder. List the teacher approval needed before cutting, drilling, soldering and first power.

**Sentence starter:** My design shows … . It meets the requirement by … .

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Use the idea

Write one measurable design goal and one constraint.
**Explain** how you will measure the goal instead of judging only by appearance.

**Sentence starter:** My design shows … . It meets the requirement by … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

An ideal series loop has supply 24 V. One component drops 4 V.
**Find** the other component’s drop.

**Word help:** series = connected along one unbranched current path

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 07 Exit check

Complete: The part that converts electrical energy into rotation is ____. Before a powered test, I will check ____.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

Propose a second success criterion that could conflict with speed, such as straight tracking or reliability.
**Describe** how both would be measured.

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** 1 2 3